SRI Quantitative Methods Bootcamp

Statistical foundations - Getting started with R

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Before We Start

1. Download files.zip: the slides, scripts, data, and RStudio project in one folder.
2. **Unzip it** (do not work inside the zip), then open **`Bootcamp.Rproj`**.
3. Open `Example-script.R` in RStudio. We will use it in the R workshop.

Install R and RStudio Desktop before the session.

Objectives of This Bootcamp

Build a common starting point for the quantitative MSc modules. By the end, you can:

- Tell a **descriptive** question from a **causal** one.
- Describe a variable: its **centre**, **spread**, and **shape**.
- Describe a **relationship** between two variables, and say what it does not show.
- Explain what a **standard error**, a **confidence interval**, and a **p-value** tell you.
- Read a simple **regression** table.
- Use **R** to import, summarise, visualise, and model data in a script you can rerun.

Plan for today

Part	The question it answers
1 · Questions and data	What can data tell us, and what are we looking at?
2 · Describing one variable	What is typical? How much do values vary?
3 · Relationships	Do two variables move together?
4 · From sample to population	How sure can we be about what we found?
5 · Regression	How much does *y* change with *x*, holding other things constant?
6 · Getting started with R	How do I do this myself?

> One question runs through the statistics: **do richer countries emit more CO2 per person?**

Questions and data

What can data tell us?

Our running example

Do richer countries emit more CO2 per person?

World Bank *World Development Indicators*, **2019**: 216 countries and economies.

Country	Region	Income group	GDP per capita (US$)	CO2 per person (t)	Population (m)
India	South Asia	Lower middle income	1,936	1.8	1389.0
Nigeria	Sub-Saharan Africa	Lower middle income	2,432	0.6	209.5
Brazil	Latin America & Caribbean	Upper middle income	8,771	2.3	207.5
China	East Asia & Pacific	Upper middle income	10,356	8.4	1407.7
United Kingdom	Europe & Central Asia	High income	47,380	5.4	66.6
United States	North America	High income	61,048	15.0	330.2
Qatar	Middle East & North Africa	High income	63,430	44.6	2.6

Two kinds of questions

Statistics helps us learn about things we cannot observe completely.

1. **Descriptive**: What is going on? *How much CO2 does a typical country emit per person?*
2. **Causal**: Why is it going on? Does X change Y? *If a country became richer, would its emissions rise?*

QUICK CHECK

Descriptive or causal?

a. Did UK emissions fall between 1990 and 2019?

b. Do carbon taxes reduce emissions?

c. Do high-income countries emit more than low-income countries?

Where the data came from

```r
# Run once, with an internet connection. The result is saved in WDI_short.RData.
library(WDI)
wd.df <- WDI(country = "all",
             indicator = c(population = "SP.POP.TOTL",          # people
                           gdp_pc = "NY.GDP.PCAP.KD",           # constant 2015 US$
                           co2_pc = "EN.GHG.CO2.PC.CE.AR5"),    # tonnes per person
             extra = TRUE, start = 2019, end = 2019)
wd.df <- wd.df[wd.df$region != "Aggregates", ]  # drop "World", "Euro area", ...
save(wd.df, file = "WDI_short.RData")
```

- `extra = TRUE` adds the **region** and **income group** of each country.
- We drop aggregates so that **each row is one country or economy**.
- Published data get revised: saving a copy lets everyone reproduce the same numbers.

Units, variables, and observations

- The **unit of analysis** is the entity we study: here, a *country or economy*. Elsewhere: people, households, schools, firms.
- A **variable** is a characteristic that varies across units: `gdp_pc`, `co2_pc`, `region`.
- An **observation** (or value) is one variable measured for one unit: *in 2019, Germany emitted 8.4 tonnes of CO2 per person.*

country	region	gdp_pc	co2_pc
Germany	Europe & Central Asia	44208	8.4
India	South Asia	1936	1.8
Nigeria	Sub-Saharan Africa	2432	0.6

> In a data table, **rows are units**, **columns are variables**, and **each cell is one value**.

Types of variables

Type	What the values are	In our data
Continuous (interval/ratio)	Numbers; calculations make sense	`gdp_pc`, `co2_pc`, `population`
Ordinal	Categories with a natural order	`income`: low → lower middle → upper middle → high
Nominal	Categories with no order	`region`

The type decides which summaries and plots make sense: an "average region" is meaningless.

QUICK CHECK

What type is: a 1–5 "strongly disagree … strongly agree" scale? Age in years? Union membership (yes/no)?

Missing values are data too

14 of 216 countries have no CO2 value; 9 have no GDP value.

- **No CO2 value:** mostly small territories (e.g. Monaco, Isle of Man), but also Serbia, Kosovo, South Sudan, and West Bank and Gaza.
- **No GDP value:** includes North Korea, Venezuela, and Eritrea.

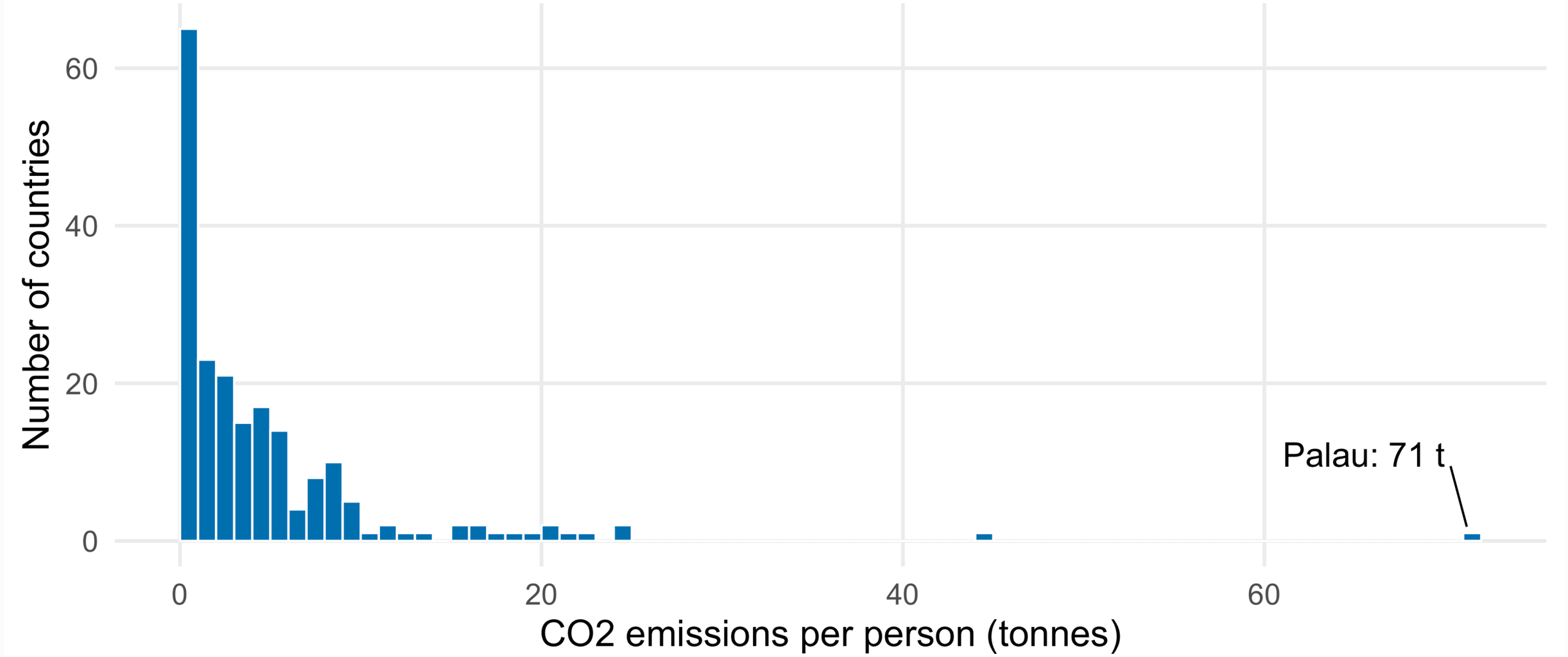
These countries are **not missing at random**: they are small, in conflict, or publish little data.

> Summaries use only the observed values. Always ask **how many** are missing, **which** ones, and **why**. Dropping them does not remove bias.

Describing one variable

What is typical? How much do values vary?

Always look at the distribution first



A **distribution** shows how often each value occurs. This one is **right-skewed**: most countries emit little; a few emit a lot.

Centre: mean and median

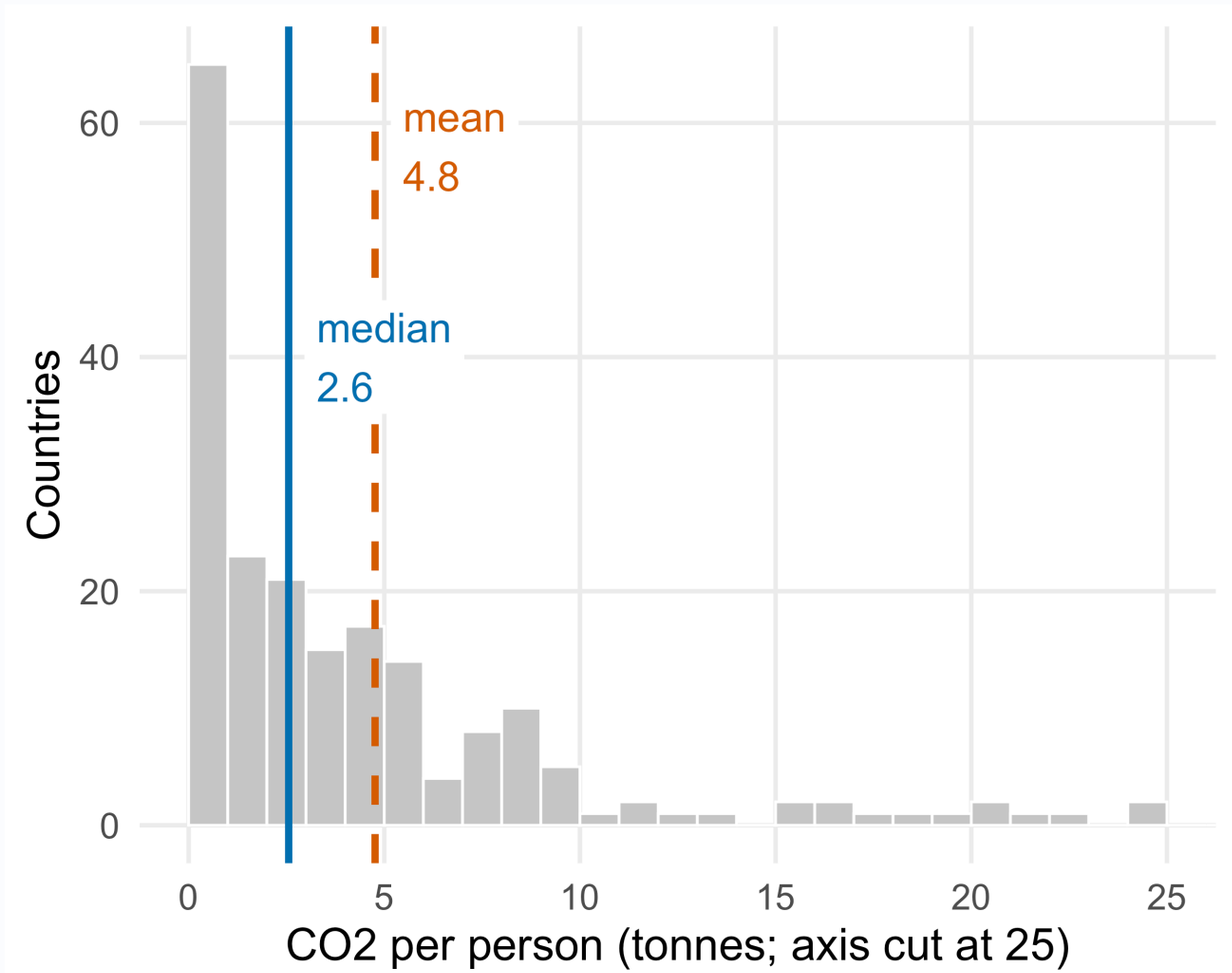
Mean: add up all values and divide by how many there are.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

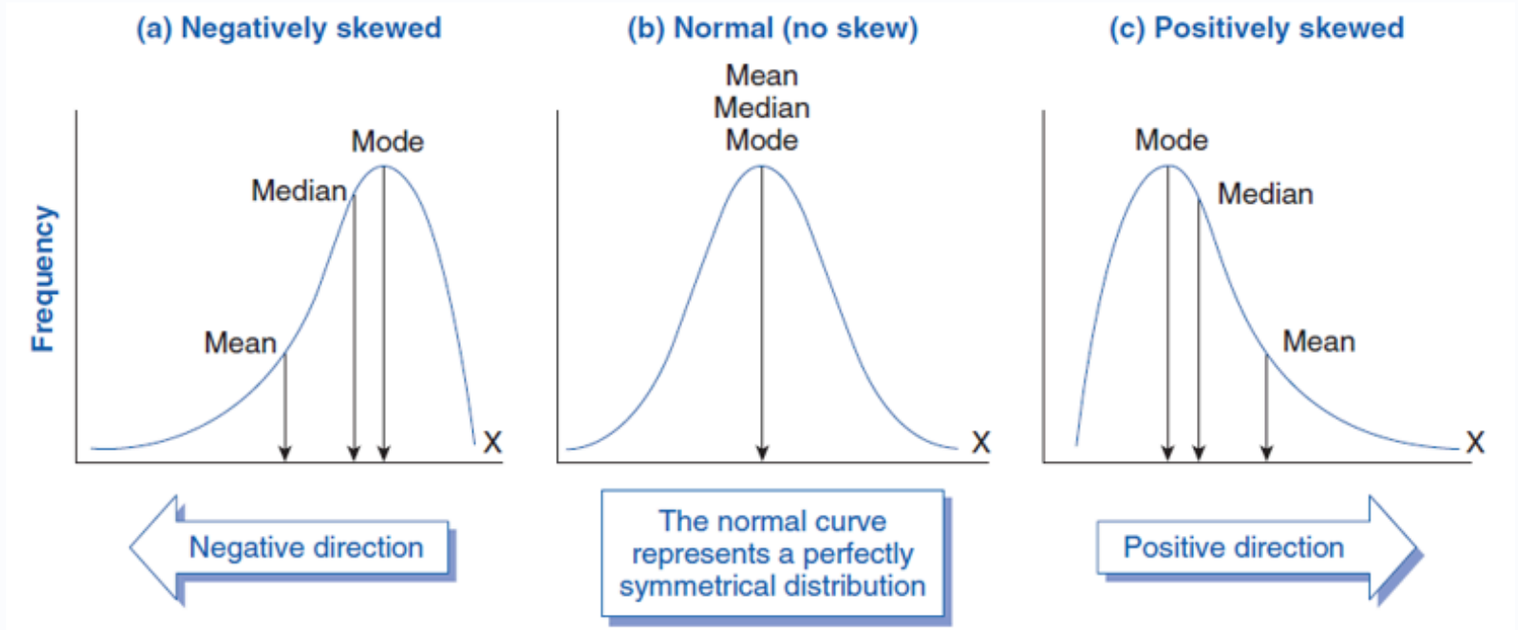
Median: the middle value once all values are sorted.

Mode: the most common value (mainly for categories).

> Mean **4.8 t**, median **2.6 t**. Half of all countries emit less than 2.6 t per person.



Why do the mean and median differ?



The mean is pulled towards the long tail; the median is not. For skewed variables such as income or emissions, report the **median** (or both).

> **QUICK CHECK**
>
> If Palau's value doubled, what would happen to the mean? To the median?

An average of countries is not an average of people

Region	Countries	Mean across countries (t)	Mean across people (t)
North America	3	12.3	15.2
Middle East & North Africa	20	10.1	5.8
Europe & Central Asia	49	6.5	7.1
East Asia & Pacific	37	6.2	6.9
Latin America & Caribbean	38	3.0	2.7
South Asia	8	1.6	1.5
Sub-Saharan Africa	47	1.0	0.8

In the Middle East & North Africa, Qatar (2.6 million people, 44.6 t) counts as much as Egypt (108 million people, 2.2 t).

> Weighting by population answers a different question: the emissions of the **average person**, not the **average country**. Decide which one your question needs.

Spread: variance and standard deviation

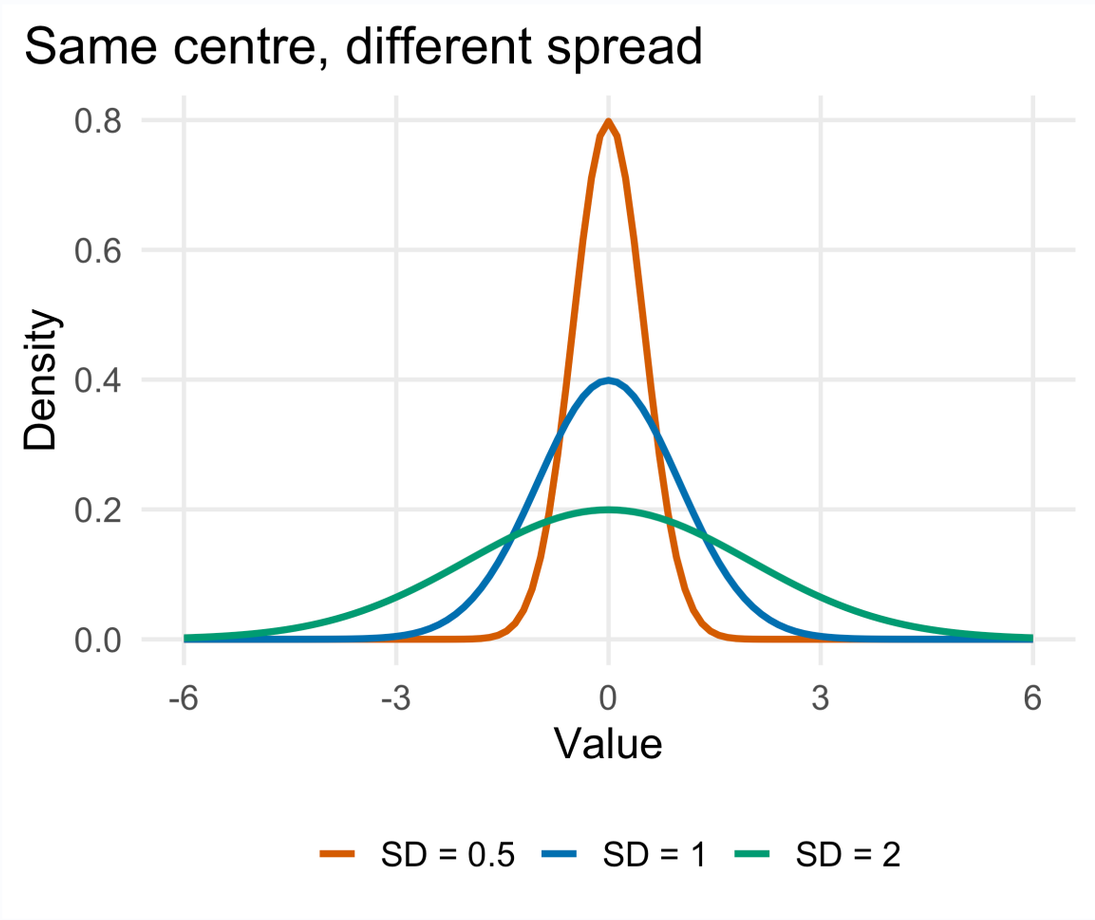
Variance: the average squared distance from the mean.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

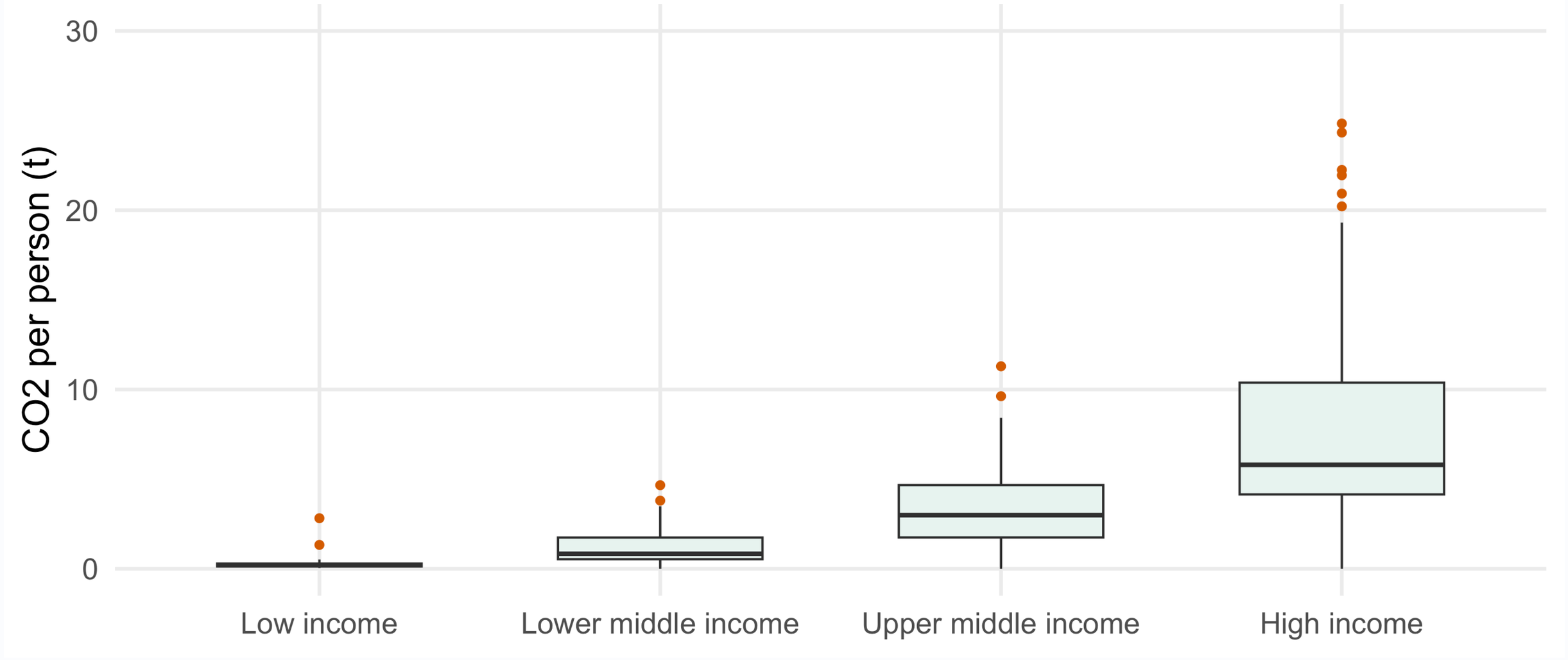
Standard deviation: the square root of the variance, back in the variable's units. Roughly, the **typical distance** of a value from the mean.

$$s = \sqrt{s^2}$$

For CO2: s = **7.4 t**, larger than the mean itself: another sign of a long tail.

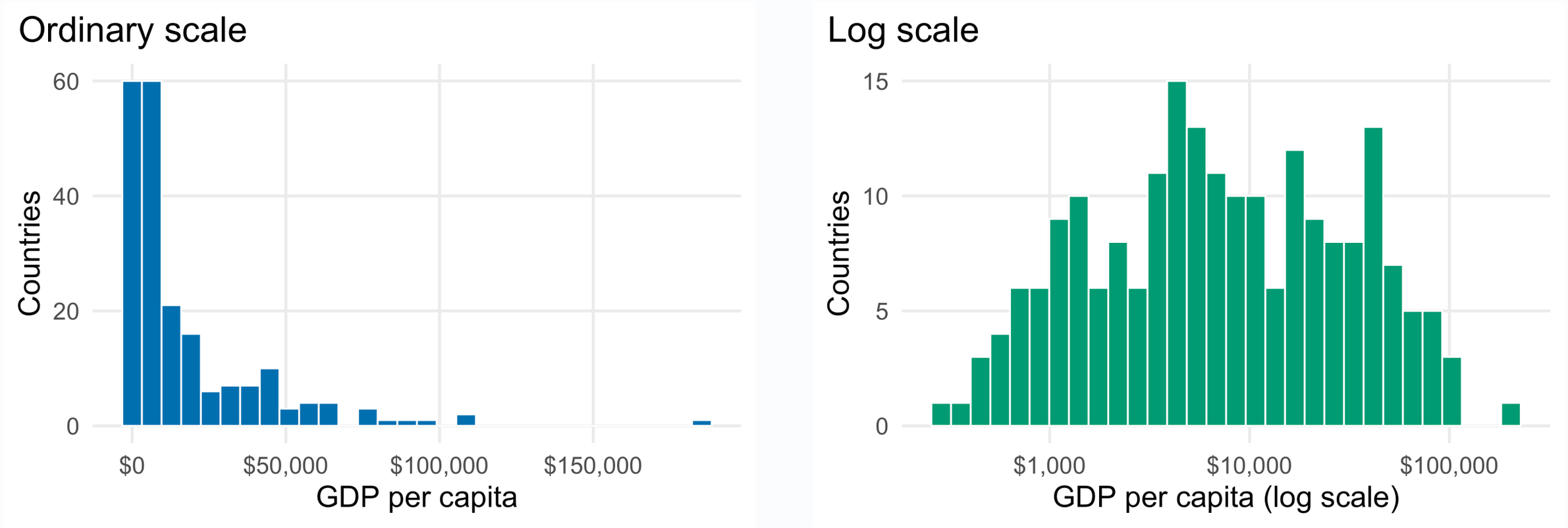


Quartiles and boxplots



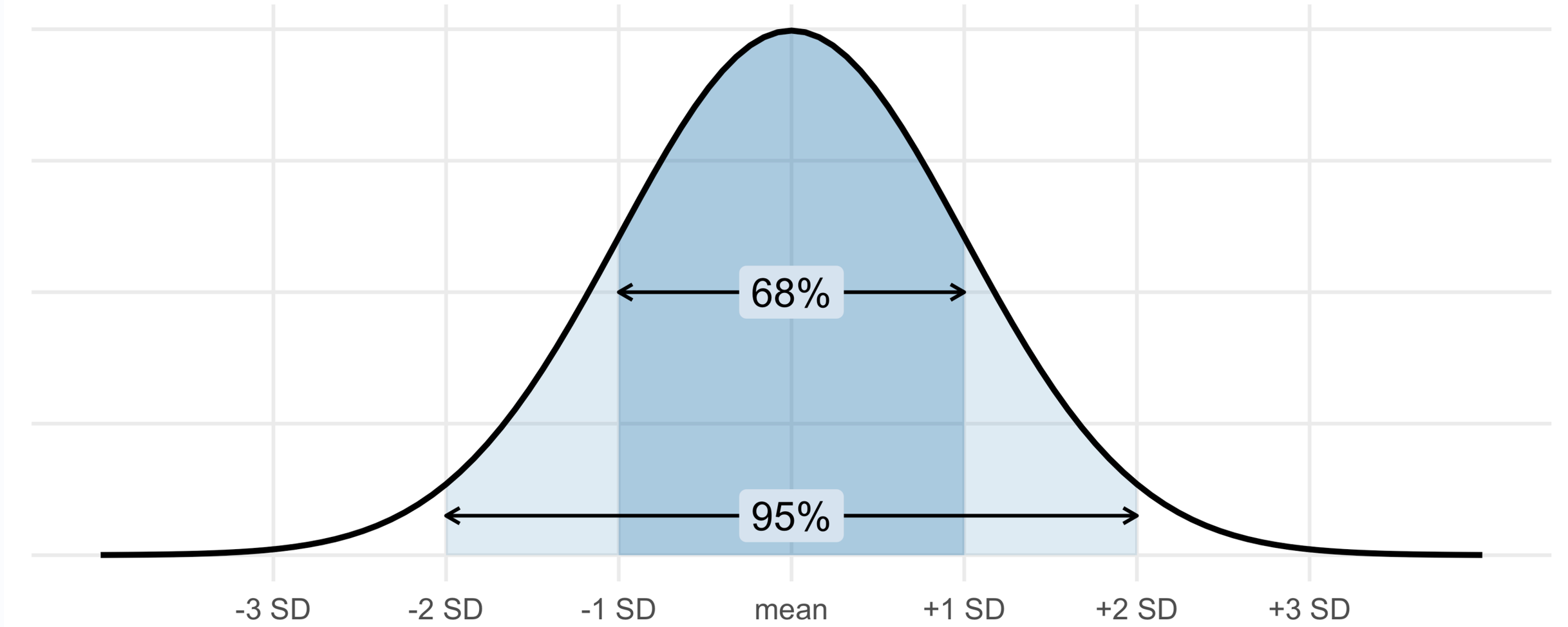
The **box** holds the middle 50% (from the 1st to the 3rd quartile), the **line** is the median, and the **dots** are unusual values. Palau and Qatar are above the axis limit of 30 t.

Skewed variables and the log scale



On a **log scale**, equal distances mean equal **ratios**: $1,000 → $10,000 → $100,000. Use it for variables that grow by multiplying, such as income, population, and emissions.

The normal distribution

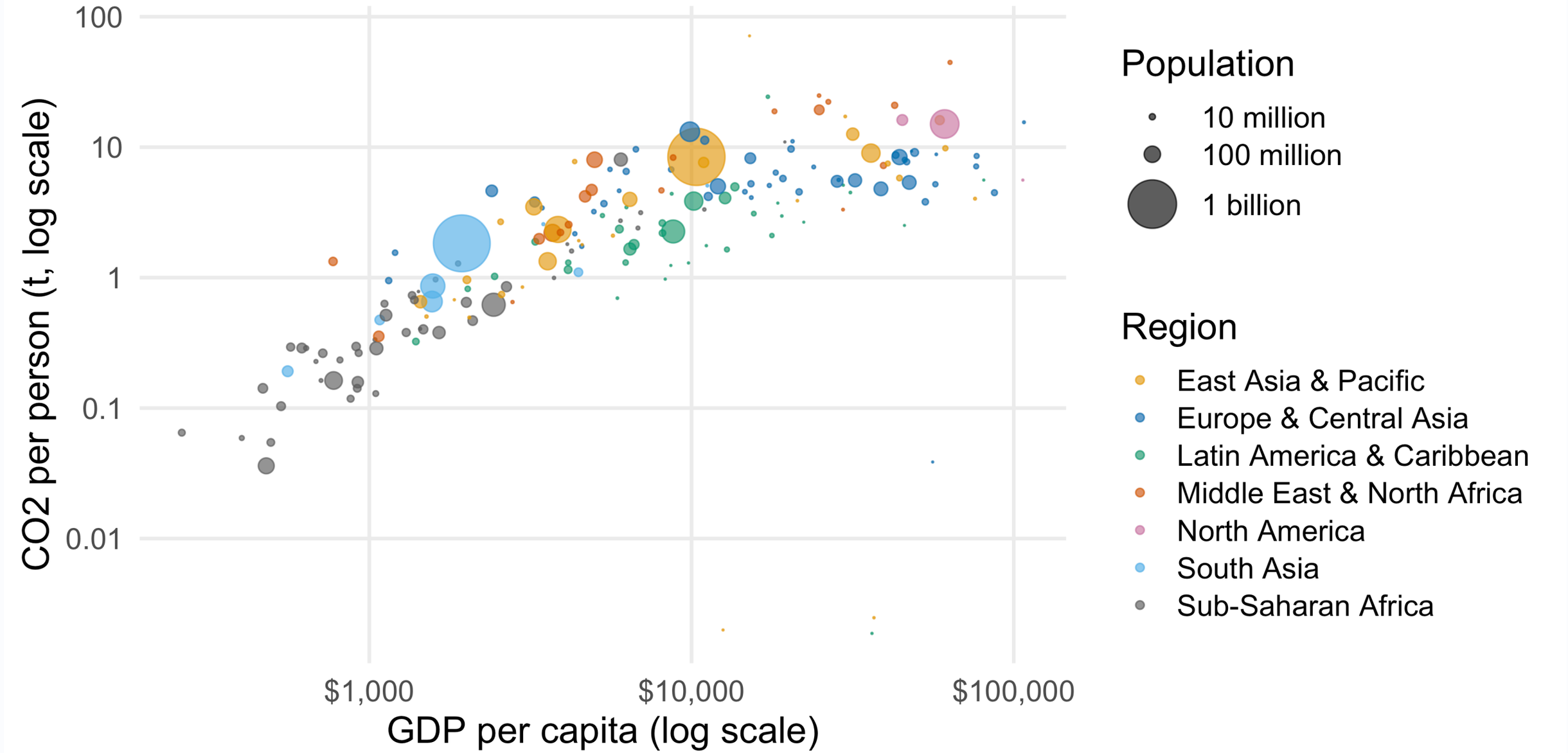


Symmetric and bell-shaped: about **68%** of values within 1 SD of the mean, **95%** within 2 SD. Few real variables look like this (CO2 does not), but **averages of samples** often do: part 4.

Relationships

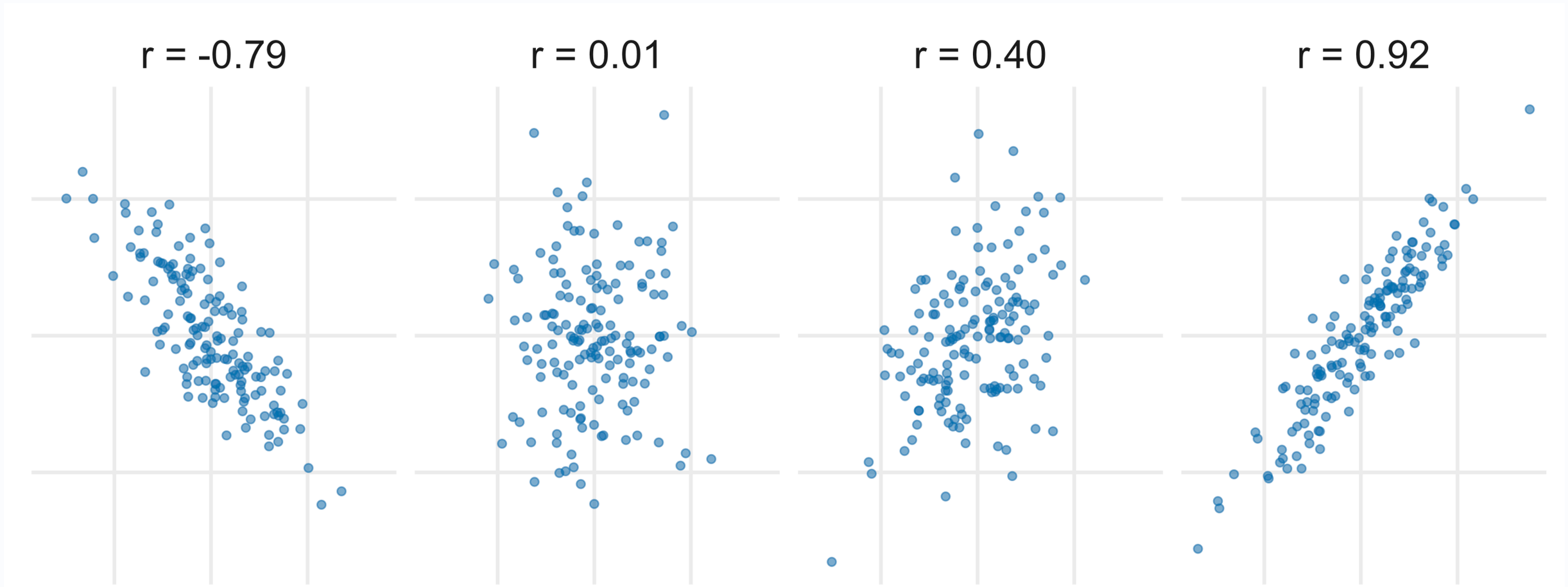
Do two variables move together?

A scatterplot shows a relationship



Each point is a country. What pattern do you see? Which countries do not fit it?

What correlations look like



Pearson's correlation r runs from **−1** (perfect negative line) through **0** (no linear relationship) to **+1** (perfect positive line). The sign gives the direction; the size gives the strength.

Pearson's correlation

Covariance: do two variables sit above (or below) their means *at the same time*?

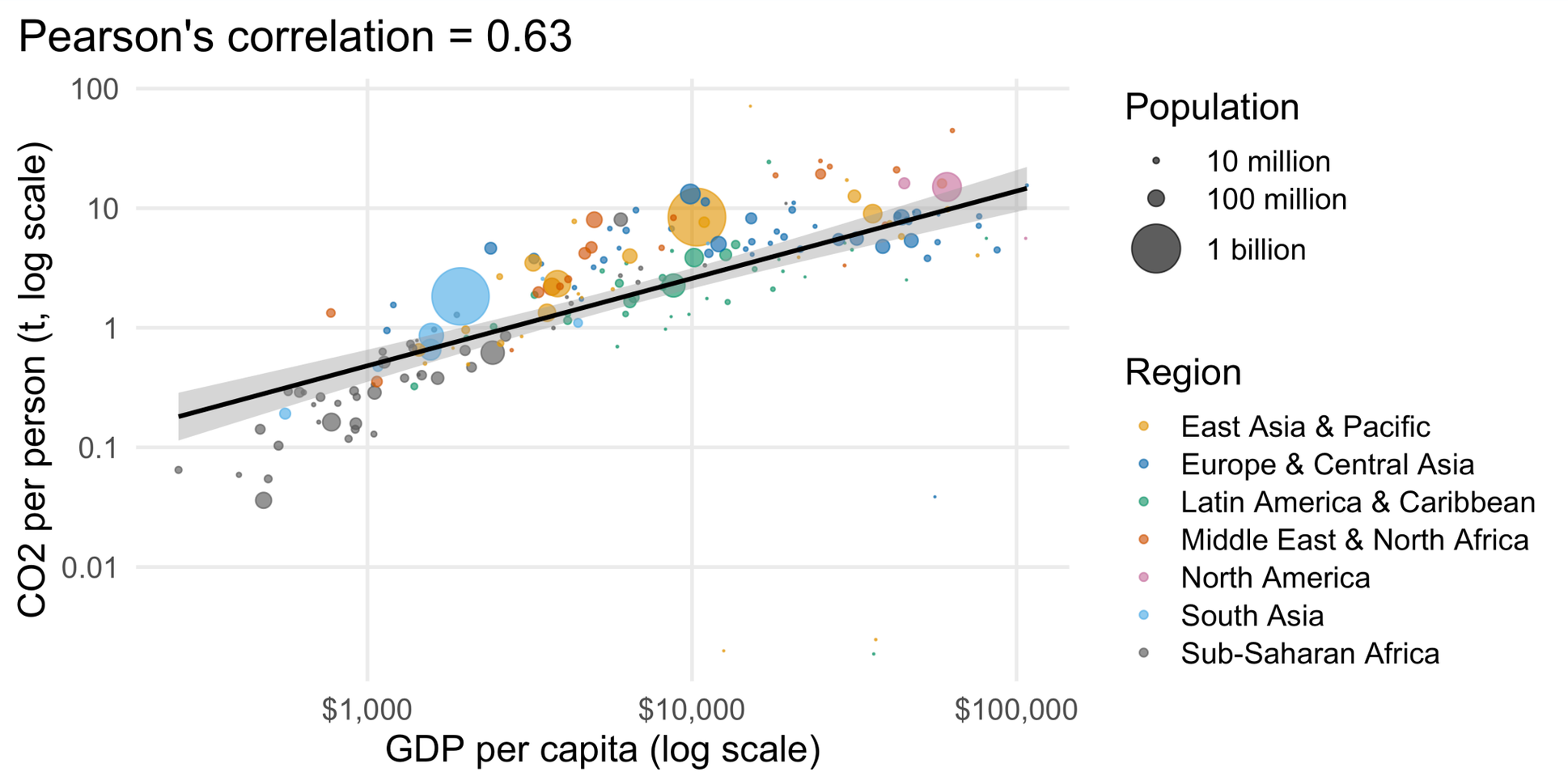
$$cov_{x,y} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Its size depends on the units (dollars? thousands of dollars?). Dividing by both standard deviations removes the units:

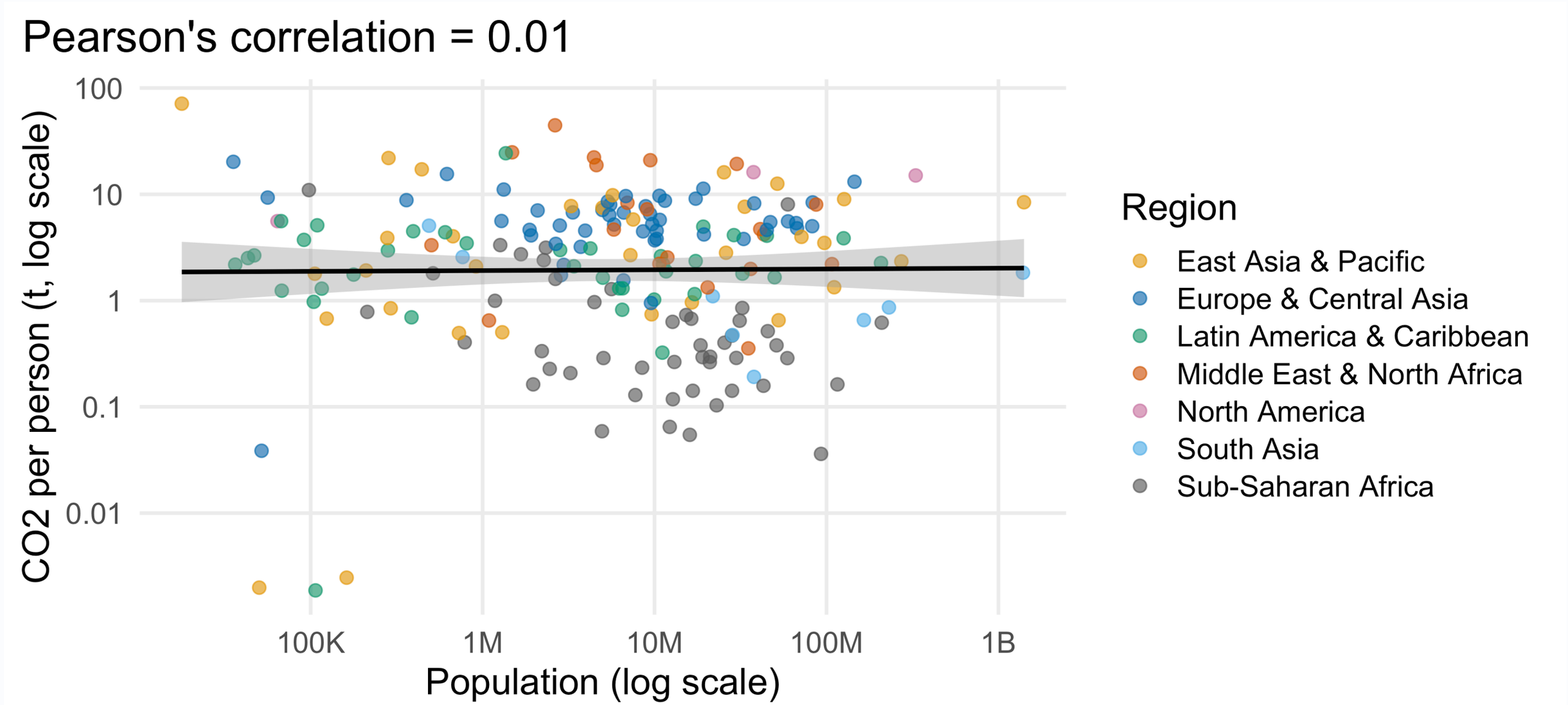
$$r = \frac{cov_{x,y}}{s_x \, s_y}$$

> r measures the strength and direction of a **linear** relationship, on a scale from −1 to +1, whatever units the variables are measured in.

Guess the correlation

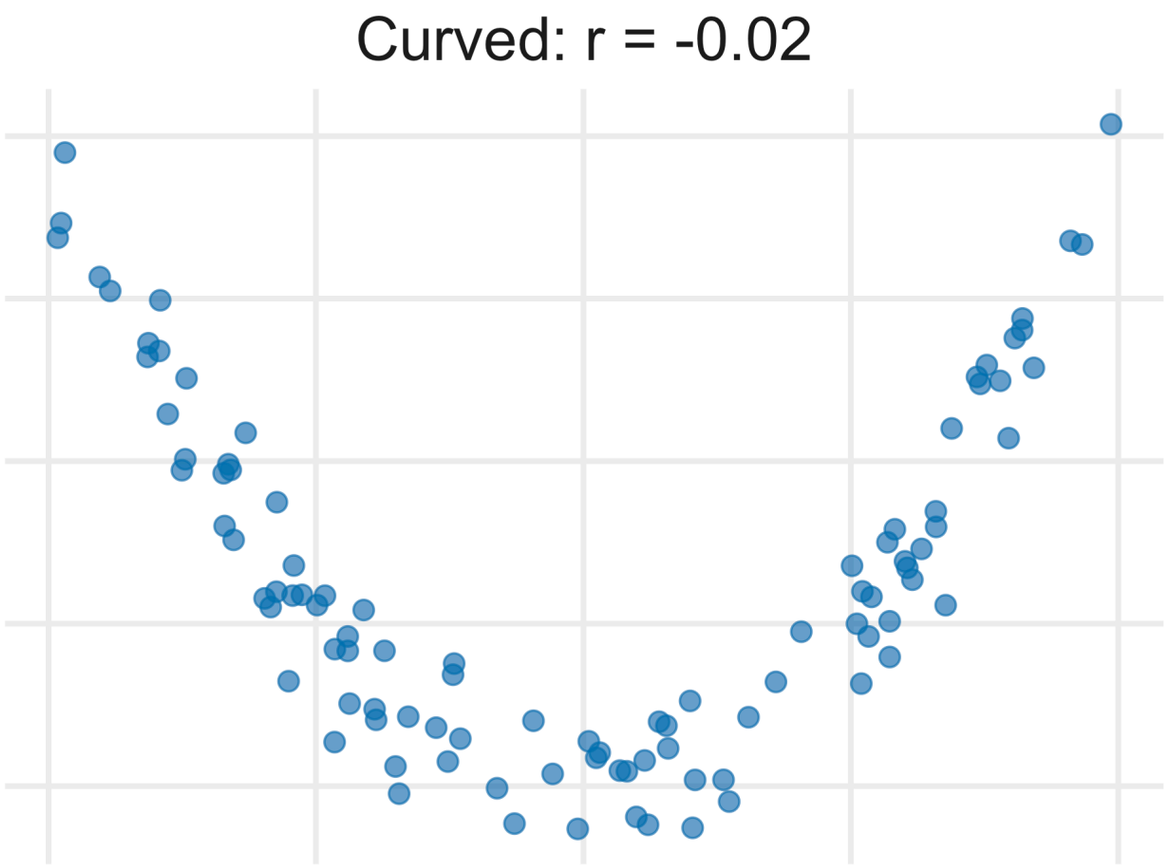


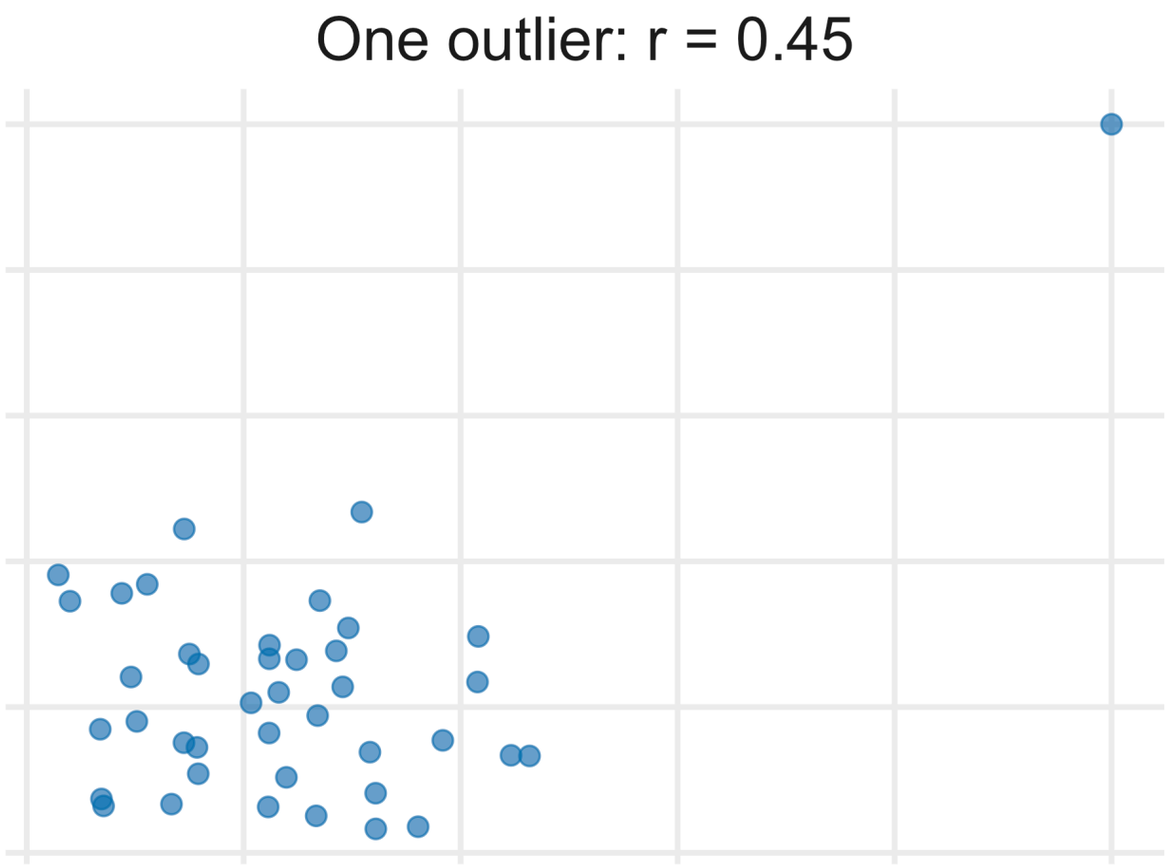
Is population related to emissions per person?



r = 0.01: knowing a country's size tells us almost nothing about its emissions per person.

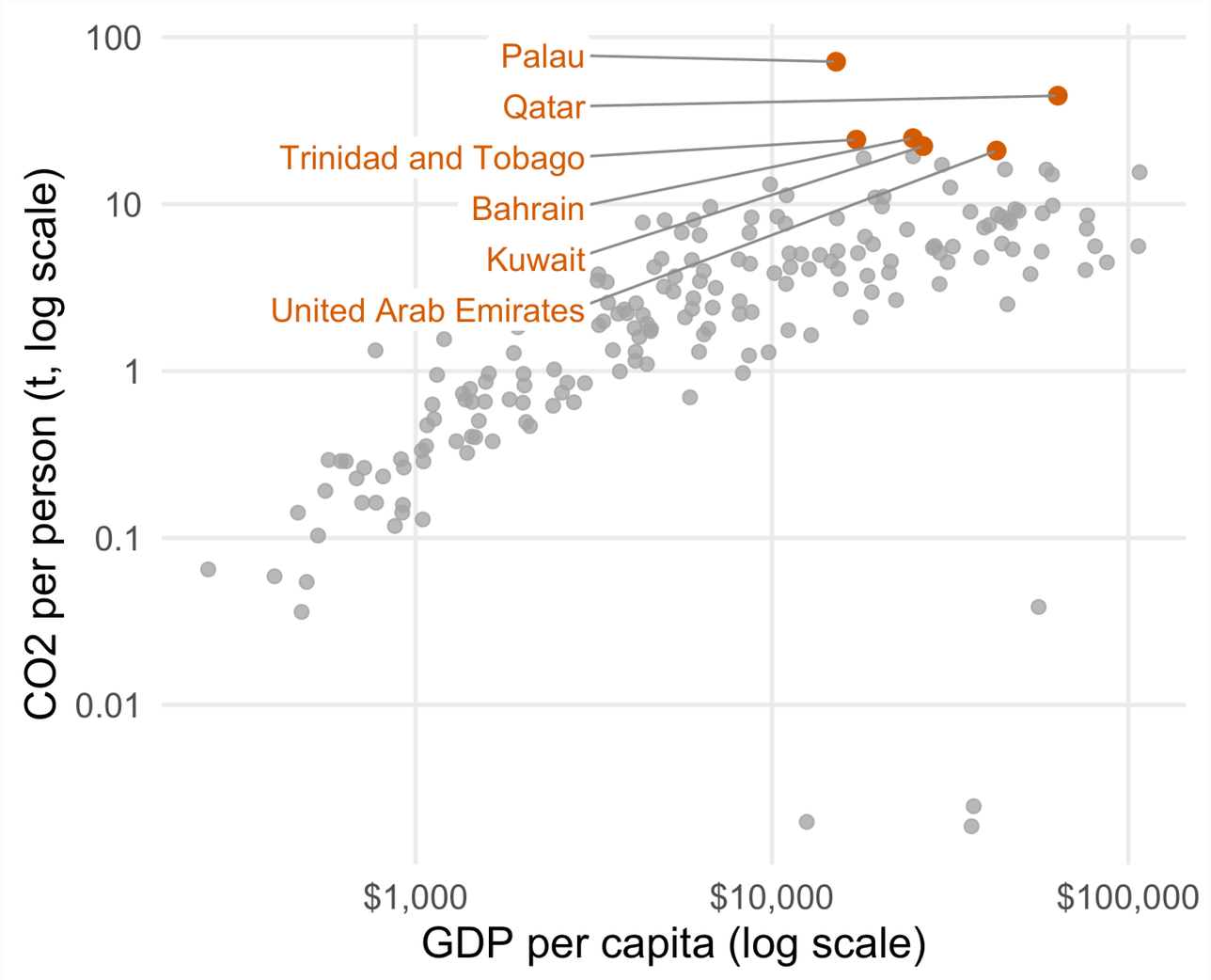
Correlation has limits



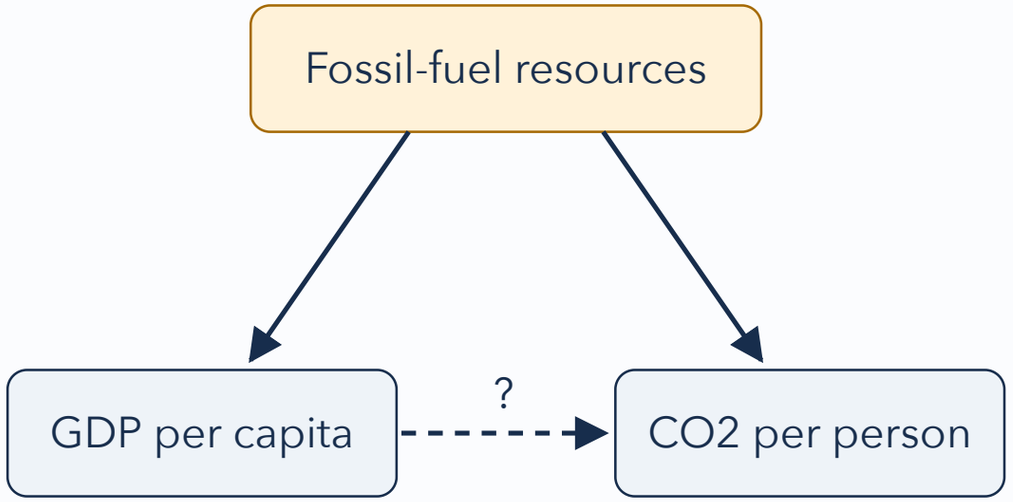


- r only captures **straight-line** relationships: a strong curve can give $r \approx 0$.
- A single extreme point can create (or hide) a correlation.
- **Plot your data** before you trust a correlation.

Correlation is not causation



Most of the top emitters are **oil and gas producers**.



A **confounder** affects both variables. It can create a correlation even if GDP itself has no effect on CO2.

Countries are not people

This plot compares **countries**. It cannot tell us whether richer **people** emit more.

Drawing conclusions about individuals from group-level data is the **ecological fallacy**. Answering questions about people needs data on people: for example, household surveys.

QUICK CHECK

"Countries with more university graduates vote more for party X, so graduates vote for party X." What is wrong with this reasoning?

From sample to population

How sure can we be?

Population and sample

- The **population** is everyone (or everything) we want to learn about. A **parameter** describes it: for example, the mean income of all UK adults.
- A **sample** is the part we observe. A **statistic** describes it and **estimates** the parameter.
- **Inference** is learning about the population from a sample, and saying how uncertain we are.

	Sample statistic	Population parameter
Mean	$\bar{x}$	μ
Standard deviation	s	σ
Correlation	r	ρ
Regression coefficient	b or $\hat{\beta}$	β

Roman letters for samples, Greek letters for populations. A hat (^) marks an estimate.

A thought experiment with our data

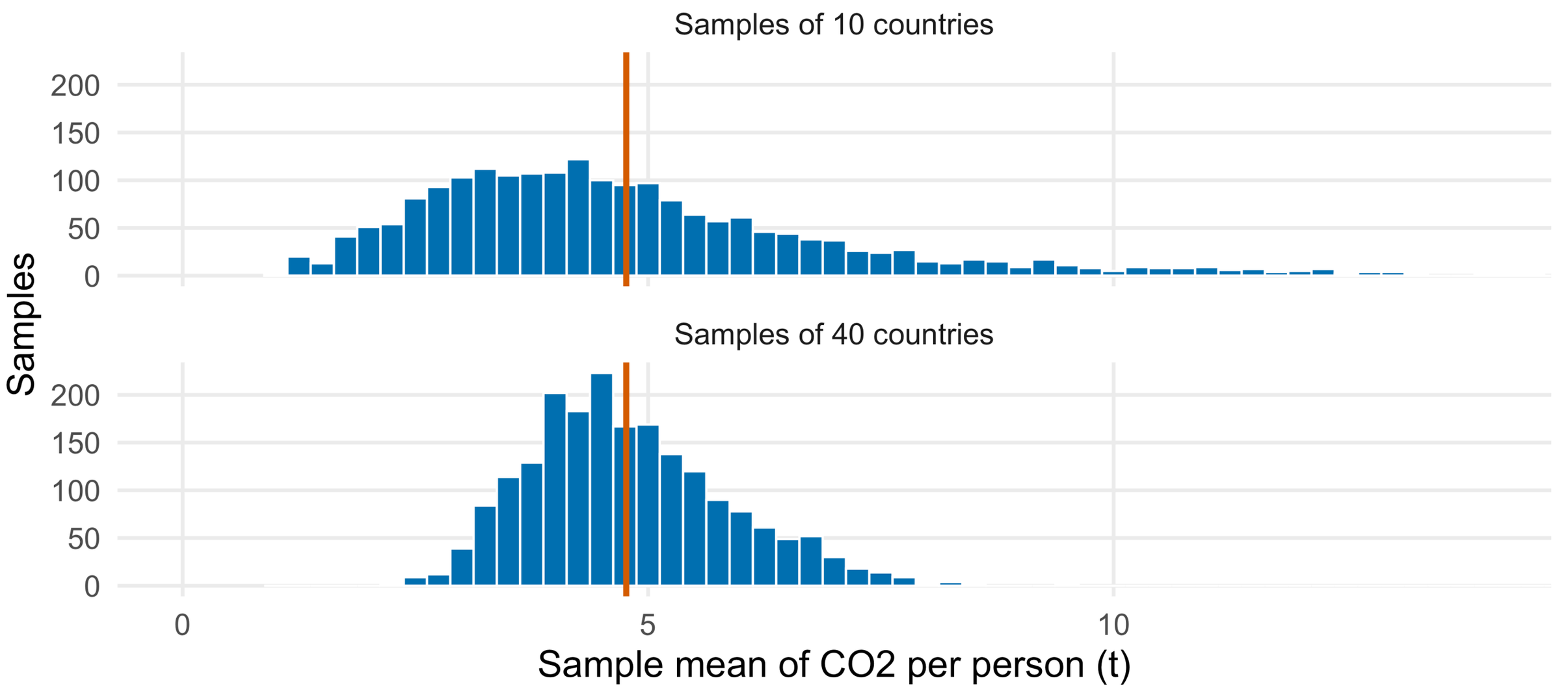
Pretend our 202 countries **are** the whole population. The true mean is μ = **4.76 t**.

Now draw a random sample of **10 countries** and compute its mean. Then do it again.

	Sample 1	Sample 2	Sample 3
Mean of 10 countries	4.25 t	3.15 t	3.51 t

> Every sample gives a different answer. This **sampling variability** is what inference measures.

The sampling distribution



2,000 samples each. Sample means **centre on the truth** (red line) and **spread less as samples grow**.

Standard error

The **standard error** (SE) is the standard deviation of the sampling distribution: how far a sample estimate typically lands from the truth.

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

	SD of the 2,000 sample means	Formula: σ / √n
Samples of 10 countries	2.32	2.35
Samples of 40 countries	1.07	1.18

> Four times the sample size halves the standard error. From **one** sample we can estimate the SE by using s in place of σ.

Confidence intervals

A **95% confidence interval** is roughly estimate ± 2 × SE.



39 of these 40 intervals contain the true mean. Across many samples, about 95% would.

Hypothesis tests and p-values

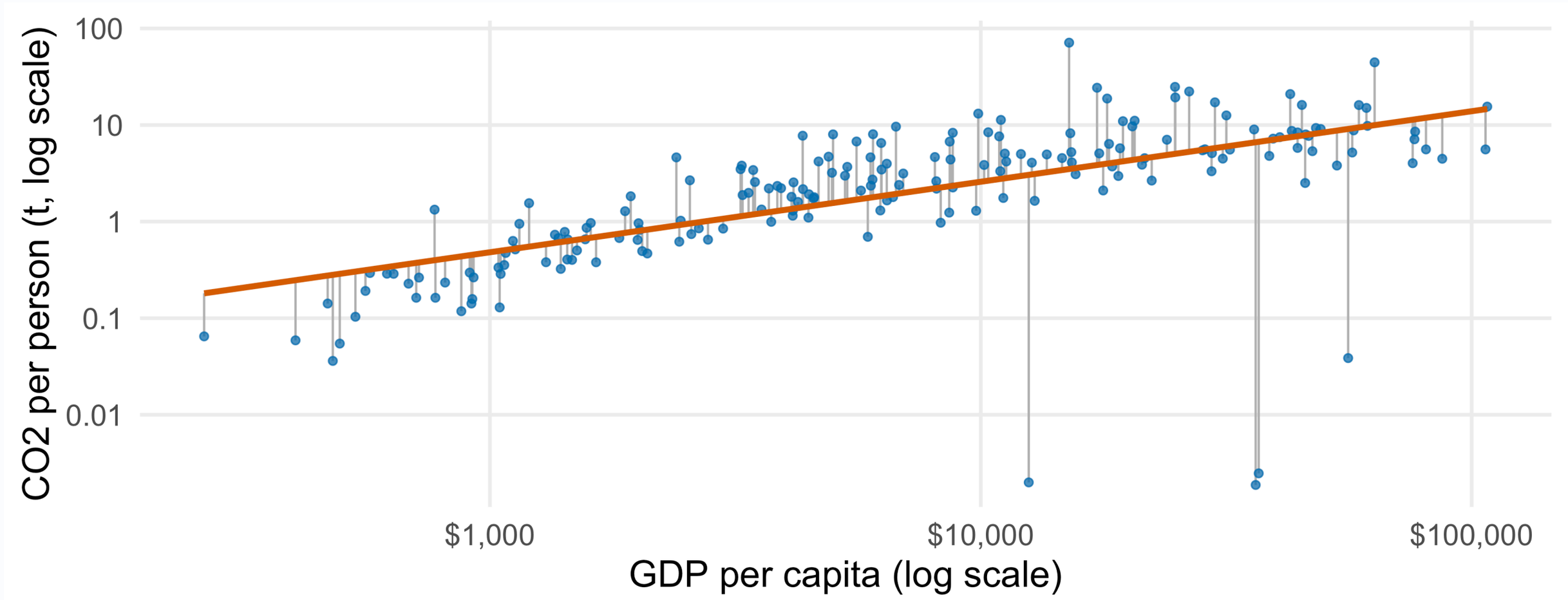
- A **hypothesis** is a theory-based expectation: *richer countries emit more CO2 per person*.
- The **null hypothesis** (H_0) says there is nothing there: *no relationship in the population*.
- The **p-value**: *if H_0 were true*, how often would sampling alone produce a result at least as extreme as ours?
- A small p-value (by convention, below 0.05) means the data would be surprising under H_0: the result is **statistically significant**.

> A p-value is **not** the probability that H_0 is true, and it says **nothing about how large or important** an effect is. Always report the estimate and its confidence interval too.

Regression

How much does *y* change with *x*?

Fitting a line



$$\hat{y_i} = b_0 + b_1 x_i \qquad \text{residual: } e_i = y_i - \hat{y_i}$$

The line predicts $\hat{y}$ from x; each grey segment is a **residual**, the prediction error.

Ordinary least squares

Ordinary least squares (OLS) chooses the intercept b_0 and slope b_1 that make the **sum of squared residuals** as small as possible:

$$\min_{b_0,\, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

The solution for the slope links straight back to correlation:

$$b_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = r \, \frac{s_y}{s_x}$$

With several explanatory variables, the model becomes $y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \epsilon_i$: same idea, more slopes. The **error term** ϵ_i holds everything that affects y but is not in the model.

Reading regression output

```
fit_co2 <- lm(log(co2_pc) ~ log(gdp_pc), data = loglog)
summary(fit_co2)
```

Term	Estimate	Std. error	95% CI	p-value
Intercept	-5.78	0.59	-6.94 to -4.62	<0.001
log(GDP per capita)	0.73	0.07	0.60 to 0.86	<0.001

- **Slope 0.73**: countries with **1% higher** GDP per capita emit, on average, about **0.73% more** CO2 per person (both variables are logged).
- **95% CI** 0.60 to 0.86: it excludes 0 ($p < 0.001$).
- **R^2 = 0.39**: GDP accounts for 39% of the variation in log CO2 across these 191 countries.

Interpreting coefficients when variables are logged

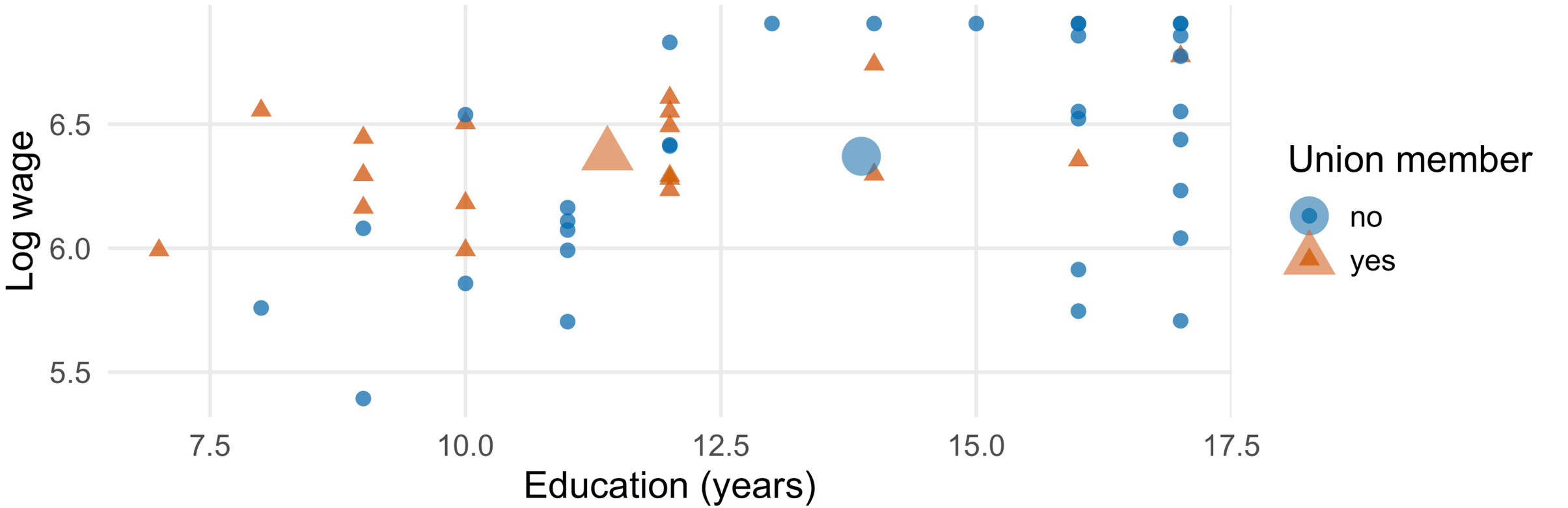
Model	A one-unit change in *x*...	Example reading of $b = 0.05$
$y = b_0 + b\,x$	changes *y* by b units	+0.05 units of *y*
$\log y = b_0 + b\,x$	changes *y* by about $100 \times b$ %	+5% in *y*
$y = b_0 + b \log x$	(1% higher *x*) changes *y* by $b/100$ units	+0.0005 units of *y*
$\log y = b_0 + b \log x$	(1% higher *x*) changes *y* by b %	+0.05% in *y*

> Keep this table: the wage model and the R workshop both use logged variables.

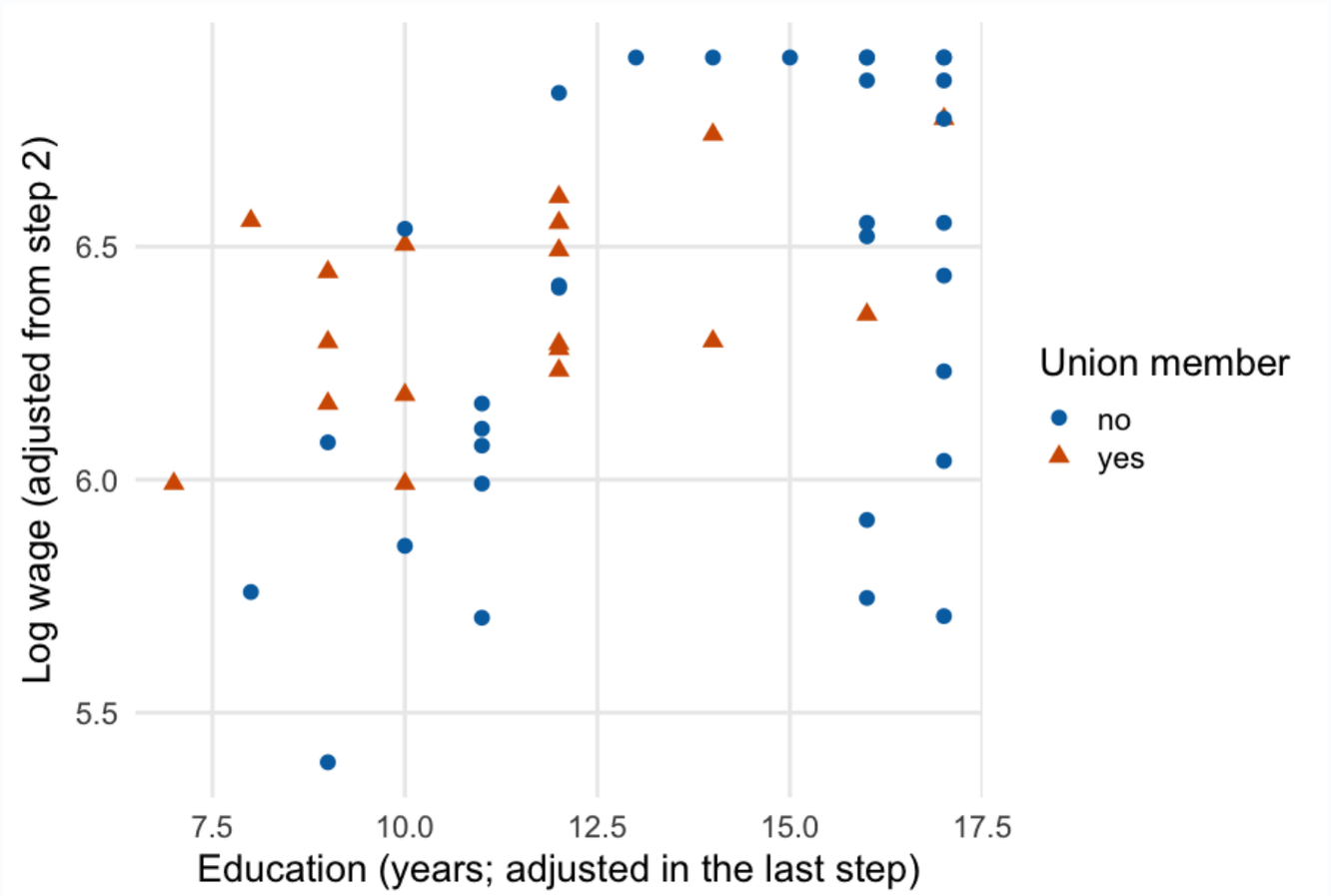
Holding other things constant

A new question, with data on **people**: how are education and union membership related to wages?

$$\log(wage_i) = \beta_0 + \beta_1\, education_i + \beta_2\, union_i + \epsilon_i$$



What “holding constant” means



1. Remove the part of wages explained by union membership.
2. Remove the part of education explained by union membership.
3. The slope between what is left is exactly b_1 in the multiple regression.

> b_1 compares people **with the same union status**.

Reading the wage model

Estimate [95% confidence interval]; outcome: log wage

Term	50 workers	All 595 workers
Education (years)	0.063 [0.028, 0.098]	0.062 [0.052, 0.073]
Union member	0.162 [-0.065, 0.388]	0.152 [0.091, 0.214]

- **Education**: one more year goes with about **6.5%** higher wages, among workers with the same union status.
- **Union**: members earn about 17.6% more than non-members with the same education.

With 50 workers the CI **includes 0** (not significant); with 595 workers it clearly does not.

> "Not significant" does not mean "no effect". Here the estimate barely changes; the sample was simply too small to estimate it precisely.

Is education's effect causal?

Probably not entirely. **Omitted variables** can affect both education and wages:

- ability and motivation
- family background and connections
- where people live and work

Adding control variables helps only for what we **measure**. Methods designed for causal questions (experiments, natural experiments, panel data) come later in your modules.

QUICK CHECK

Name one variable, not in this model, that might affect both education and wages. Which way would leaving it out bias the education coefficient?

Key takeaways

1. Start with the question: **descriptive** or **causal**? Then look at the data.
2. Describe distributions with **centre, spread, and shape**. Skew makes the median more useful.
3. Averages of countries are not averages of people; country patterns are not individual patterns.
4. Correlation measures **linear association**, not causation. Watch for confounders.
5. Estimates vary from sample to sample. **Standard errors** and **confidence intervals** say by how much.
6. A regression coefficient describes a relationship **holding the other variables in the model constant**.

Words to know

Term	In one line
Unit, variable, observation	What we study; what we measure; one measurement
Distribution	How often each value occurs
Mean / median	Average / middle value
Standard deviation	Typical distance from the mean
Correlation (r)	Strength and direction of a linear relationship, −1 to +1
Confounder	A variable that affects both *x* and *y*
Standard error	How much an estimate varies from sample to sample
95% confidence interval	Estimate ± about 2 SE; captures the truth in 95% of samples
p-value	How surprising the data would be if the null hypothesis were true
Regression coefficient	Expected change in *y* for a one-unit change in *x*, others held constant

Getting started with R

From a question to a reproducible result.

One small analysis, start to finish

R WORKSHOP

How do GDP per capita and life expectancy vary across countries?

Import | Inspect | Transform | Summarise | Visualise | Model

By the end, you can run an R script, explore a dataset, and save a result.

> Work along in `Example-script.R`. All examples use `WDI_Data.csv`.

R + RStudio: two parts of your setup

01 · R

The language that runs your calculations and creates your plots.

Install R from CRAN **first**.

02 · RStudio

The workspace where you write scripts, inspect data, and view results.

Then install RStudio Desktop.

> Open RStudio. In the Console, run `1 + 1`. A result of `2` means R is working.

Know where your code lives

Source · write

Save your steps in an `.R` script.

```
minutes <- 3 * 60
```

Environment · inspect

See the objects in this session.

```
minutes → 180
```

Console · run

R executes code and prints results.

```
[1] 180
```

Files / Plots / Help · explore

Browse files, see figures, and read function documentation.

Open `Example-script.R`. Run a line or selection with **Ctrl + Enter** (Windows/Linux) or **Cmd + Enter** (Mac). Pane positions can vary.

Keep the project together

```
files/
  Bootcamp.Rproj
  Example-script.R
  R-workshop.R
  WDI_Data.csv
  Readme.txt
  SRI Quantitative Methods Bootcamp.pdf
  outputs/
```

1. Unzip `files.zip`. Keep the files together.
2. Open **`Bootcamp.Rproj`** in RStudio.
3. Open your script and save your edits.

File paths now start from this folder.

> Use `"WDI_Data.csv"` in your code. Use `getwd()` and `list.files()` to check where R is looking.

Objects give results a name

```
hours <- 3
minutes <- hours * 60
minutes
```

```
[1] 180
```

- `<-` assigns a value to a name; typing the name prints its value.
- `#` starts a comment: use it to explain your choices.
- Names are case-sensitive: `minutes` and `Minutes` are different.

> Change `hours` to `4`. Rerun all three lines. What changes?

Vectors hold several values

```
scores <- c(60, 70, 80, NA)
scores[1]           # first element
```

```
[1] 60
```

```
scores >= 70        # a logical test
```

```
[1] FALSE  TRUE  TRUE    NA
```

```
length(scores)      # number of elements
```

```
[1] 4
```

`c()` combines values. `[1]` selects the first element. `NA` marks a missing value.

Numeric: `c(60, 70)` · Character: `c("Asia", "Europe")` · Logical: `c(TRUE, FALSE)`

Functions do things to objects

```
mean(scores)                # NA: a value is missing
```

```
[1] NA
```

```
mean(scores, na.rm = TRUE)   # mean of observed scores
```

```
[1] 70
```

```
sum(is.na(scores))          # how many are missing?
```

```
[1] 1
```

Read `mean(scores, na.rm = TRUE)` as: **function** + **data** + **argument controlling missing values**.

> Your turn · Calculate the median of the observed scores.

Packages: install once, load each session

Run this in the Console **once** if the packages are missing:

```
install.packages(c("readr", "dplyr", "ggplot2"))
```

Keep these lines at the top of your script:

```
library(readr)    # import and export data
library(dplyr)    # transform and summarise
library(ggplot2)  # visualise
```

> R already includes `mean()`, `median()`, and `sum()`. Packages add more tools.

Import your first dataset

```
wdi <- read_csv("WDI_Data.csv", show_col_types = FALSE)
dim(wdi)  # rows, then columns
```

```
[1] 217  61
```

A **data frame** is a table: rows are observations; columns are variables.

Our data	What it means
One row	A country or economy in the supplied WDI extract
`gdpcap`	GDP per capita, constant 2015 US dollars
`lifexp_tot`	Life expectancy at birth, years

Inspect before you analyse

```
wdi |>
  select(countryname, continent, gdpcap, lifexp_tot) |>
  head(3)
```

```
# A tibble: 3 x 4
  countryname continent gdpcap lifexp_tot
  <chr>       <chr>      <dbl>      <dbl>
1 Afghanistan Asia        555.       64.8
2 Albania     Europe     4543.       78.6
3 Algeria     Africa     4115.       76.9
```

Use `$` to extract a column as a vector:

```
mean(wdi$gdpcap, na.rm = TRUE)
```

```
[1] 16782.43
```

`names(wdi)` lists variables; `glimpse(wdi)` shows their types; `View(wdi)` opens a table in RStudio.

Read a pipeline from top to bottom

```
europe <- wdi |>
  filter(continent == "Europe") |>
  select(countryname, gdpcap, lifexp_tot)
```

1. Start with `wdi`.
2. `filter()` keeps **rows** that match a condition.
3. `select()` keeps the **columns** you need.

> |> passes a result into the next function. == tests equality; <– assigns.

Make a variable; choose your observations

```
plot_data <- wdi |>
  filter(!is.na(gdpcap), !is.na(lifexp_tot), gdpcap > 0) |>
  mutate(gdp_thousands = gdpcap / 1000)
```

- `is.na()` identifies missing values; `!` means “not”.
- Conditions separated by commas in `filter()` must **all** hold.
- `mutate()` adds or changes columns.

> We will plot complete GDP/life-expectancy pairs. A log axis also needs positive GDP values.

Summarise by group

```
gdp_by_continent <- wdi |>
  group_by(continent) |>
  summarise(
    n_gdp = sum(!is.na(gdpcap)),
    mean_gdp = mean(gdpcap, na.rm = TRUE),
    .groups = "drop"
  )
gdp_by_continent
```

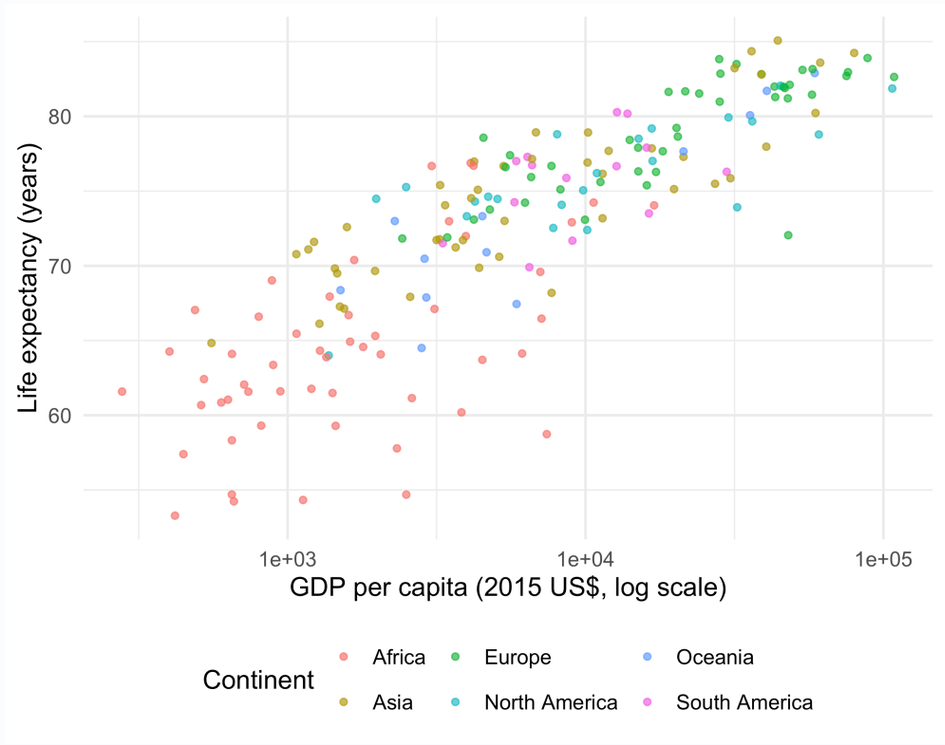
Continent	n	Mean GDP
Africa	52	2610
Asia	46	14225
Europe	47	34863
North America	26	22296
Oceania	18	13552
South America	15	11420

Count the observations used. `n_gdp` counts non-missing GDP values.

Each country/economy has equal weight. This is not a population-weighted mean.

Build a plot in layers

```
gdp_plot <- ggplot(plot_data,
    aes(x = gdpcap, y = lifexp_tot,
        colour = continent)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  labs(x = "GDP per capita (2015 US$, log scale)",
       y = "Life expectancy (years)",
       colour = "Continent") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "bottom")
print(gdp_plot)
```



`aes()` maps variables; `geom_point()` draws points; `+` adds layers.

Describe the association. Does this plot establish causation?

From plot to model

```
life_model <- lm(lifexp_tot ~ log(gdpcap), data = plot_data)
coef(life_model)     # intercept and slope
```

```
(Intercept) log(gdpcap)
  33.996598    4.448503
```

```
confint(life_model)  # 95% confidence intervals
```

```
                2.5 %    97.5 %
(Intercept) 30.416794 37.576402
log(gdpcap)  4.044084  4.852922
```

`lm(outcome ~ explanatory, data = ...)` fits a linear regression, as in the statistics part.

The slope is **4.45**: a country with **double** the GDP per capita has, on average, about **3.1 more years** of life expectancy (4.45 × log 2).

> `summary(life_model)` adds standard errors, p-values, and R^2. Association is not causation.

Your turn: a five-minute investigation

TRY → COMPARE → EXPLAIN

1. Keep countries/economies in **Europe**.
2. Calculate **mean life expectancy** and the **number with observed life expectancy**.
3. Adapt your code to compare **all continents**.

> Use `filter()`, `group_by()`, `summarise()`, `mean()`, and `sum(!is.na(...))`.

Explain your result in one sentence. What does it average over?

Save the result; make it rerunnable

```
dir.create("outputs", showWarnings = FALSE)
write_csv(gdp_by_continent, "outputs/gdp_by_continent.csv")
ggsave("outputs/gdp_life_expectancy.png", gdp_plot,
       width = 8, height = 5, dpi = 150)
```

Save your script → Session > Restart R → run from the top.

> Keep raw data, code, and outputs together. Quarto can later combine your code, results, and explanation in one report.

When you get stuck

What you see	First thing to check
`object 'wdi' not found`	Run the import line; check spelling and capitals.
`could not find function`	Load the package with `library()`.
`No such file or directory`	Check `getwd()` and `list.files()`.
A lone + in the Console	Finish the bracket/quote, or press **Esc**.

Read the first error. Try `?mean` or `?dplyr::filter` for examples and arguments.

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